Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech.
Code: 100308

Subject: Electromagnetic Fields

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer of the following (any seven question only):*** **[2 x 7 = 14]**

(a) Intrinsic impedance is

(i) $\sqrt{\frac{j\omega}{\sigma + j\omega\varepsilon}}$ (ii) $\sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$

(iii) $\sqrt{\frac{j\omega\varepsilon}{\sigma + j\omega\mu}}$ (iv) None of the above

(b) Attenuation constant α for lossless dielectric is given by

(i) $\sqrt{\pi f \mu \sigma}$ (ii) 1 (iii) ∞ (iv) 0

(c) Loss tangent angle foe perfect dielectric should be

(i) Minimum (ii) maximum
(iii) Infinity (iv) None of the above

(d) Let us consider wave is propagating along positive z-direction then direction $\vec{E}$-field & $\vec{H}$-field will be

(i) $\vec{E}$-field along $\widehat{a_x}$; $\vec{H}$-field along $\widehat{a_y}$ (ii) $\vec{E}$-field along $\widehat{a_y}$; $\vec{H}$-field along $\widehat{a_x}$

(iii) $\vec{E}$-field along $\widehat{a_z}$; $\vec{H}$-field along $\widehat{a_x}$ (iv) None of the above

(e) Which of the following is the reason for strong magnetism in a ferro magnetism?

(i) Magnetic susceptibility (ii) Magnetic permittivity
(iii) Magnetic Di-polarity (iv) Both (i) and (iii)

(f) The intrinsic impedance of free space is

(i) 277Ω (ii) 265Ω
(iii) 377Ω (iv) 177Ω

(g) For a plane wave propagating in an unbounded medium (say, free space), the minimum angle between electric field and magnetic field vectors is

(i) 0° (ii) 60°
(iii) 90° (iv) 180°

(h) The gradient of $\vec{A} = x\,i + y\,j + z\,k$ is

(i) 0 (ii) 2
(iii) 3 (iv) 1

(i) The line integral of magnetic field intensity is

(i) current (ii) current Density
(iii) μI (iv) none

(j) Maxwell's fourth equation gives .....

(i) $\nabla . \vec{E} = \frac{-\partial \vec{B}}{\partial t}$ (ii) $\nabla \times \vec{E} = \frac{-\partial \vec{B}}{\partial t}$

(iii) $\nabla . \vec{E} = \frac{\partial \vec{B}}{\partial t}$ (iv) $\nabla \times \vec{E} = \frac{\partial \vec{B}}{\partial t}$

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Q.2 (a) Convert a rectangular point P(-2,6,3) into cylindrical & spherical co-ordinate system. [7]

(b) $\vec{E}$ and $\vec{H}$ waves, travelling in free space, are normally incident on the interface with perfect dielectric with $\varepsilon_r = 3$. Compute the magnitudes of incident, reflected and transmitted $\vec{E}$ and $\vec{H}$ waves at the surfaces. [7]

Q.3 (a) Explain Convection current, Conduction current and Displacement current. Also derive convection & conduction current density. [10]

(b) Write conversion matrix for a vector from Cartesian coordinate to spherical coordinate system. [4]

Q.4 (a) Find the Divergence of vector $\vec{D}$,where [7]

$$\vec{D} = \frac{1}{r^2}\cos\theta\ \hat{a}_r + r\sin\theta\cos\emptyset\hat{a}_\theta + \cos\theta\hat{a}_\emptyset$$

(b) State and explain Gauss's law. Also write its application. [7]

Q.5 (a) Derive the boundary condition of the normal and tangential components of magnetic field at the interface of two medium with different dielectrics. [8]

(b) Given $\vec{E} = Em\ \text{Sin}(\omega t - \beta z)\ \hat{a}_y$ in free space .find $\vec{D}$, $\vec{B}$ and $\vec{H}$. sketch $\vec{E}$ and $\vec{H}$ at t=0. [6]

Q.6 (a) Explain the application of Ampere's law for finding magnetic field in case of hollow and solid conductor carrying current "I". https://www.akubihar.com [7]

(b) What is transformer emf ? Show that $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$. [7]

Q.7 (a) Write Maxwell's four equation in differential and integral form and its physical meaning. [7]

(b) Explain transformer and motional emf in detail. [7]

Q.8 A medium like copper conductor which is characterized by the parameters $\sigma = 5.8 \times 10^7$ mho/m, $\varepsilon_r = 1$, $\mu_r = 1$ support a uniform plane wave of frequency 60 Hz. Find the attenuation constant ,propagation constant , intrinsic impedance, wavelength and phase velocity of the wave. [14]

Q.9 Write short notes on ***any two*** of the following: [7x2=14]

(a) Skin depth

(b) uniqueness theorem

(c) Explain magnetization of material

(d) Biot-savert law.

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